From Connectivity to Intelligence: UAE Capacity Building Week

Program and Learning Guide

19–22 October 2026 | TDRA headquarters, Dubai

Delivered by the International Telecommunication Union (ITU) in cooperation with TDRA

About the program

The training is part of setting up the ITU AI for Good Sandbox in the UAE. The Sandbox is a shared, standards-based environment where government teams, researchers and startups can develop, test and validate AI solutions safely before they are put into service.

The week brings together an opening ceremony with a high-level panel and a training program in three levels. Each training level is designed for a different role, and each build on the one before it. Participants join the level that matches their role and experience.

Session	Date	Duration	Who it is for	Focus
1. Basic	Monday 19 October 2026	10:00 am – 01:00 pm	Government employees at all levels, youth, diplomats and anyone new to AI	How AI and machine learning work
2. Advanced	Tuesday 20 October 2026	10:00 am – 03:00 pm	AI specialists, technical teams and experts in vital sectors	Analyzing and building AI solutions in the Sandbox
Opening Ceremony and High-Level Panel	Wednesday 21 October 2026	9:00 am – 04:00 pm	Their Excellencies or their representatives, by personal invitation	Official opening, AI strategies, policies and announcements
3. Leaders	Thursday 22 October 2026	10:00 am – 12:00 pm	Leaders and decision makers (Their Excellencies or their representatives)	Leading AI adoption: strategy, governance and agentic AI in government

International standards behind the training

The training follows ITU-T Recommendations on AI and machine learning:

- Y.3172: the standard pipeline for building machine learning systems
- Y.3174: handling data for machine learning
- Y.3176: AI and machine learning marketplaces
- Y.3179: preparing and serving AI models in different environments
- Y.3181: the architecture of the AI and machine learning sandbox
- Y.3187: end-to-end orchestration of AI solutions

Level 1: Basic Training in AI and Machine Learning

Date: Monday 19 October 2026

Duration: 10:00 am – 01:00 pm

Who it is for: Government employees at all levels, youth, diplomats and anyone who wants a solid foundation in AI. No technical background is needed.

This level explains how AI systems are built, from the data that goes in to the service that comes out. Each step is explained with examples from real AI projects, and participants practice with use cases in class.

What you will learn

1. **How an AI system works:** the basic vocabulary of AI and the standard machine learning pipeline (ITU-T Y.3172). The pipeline follows data from its source through collection, preparation, the model, the rules that govern its use, and delivery to the user.
2. **Data: the raw material of AI:** the types and sources of data used in AI, including text, images, audio, video, numbers and networks of connected data, with examples from ITU open datasets.
3. **Where AI runs: devices or the cloud:** the trade-offs between running AI on a device and in the cloud, covering speed, size, energy use and cost. Also covers cleaning and anonymizing data, and encryption to protect privacy.
4. **Training and testing AI models:** how models learn from labelled data and human supervision, how data is split for training and testing, and how performance is measured against industry benchmarks.
5. **Types of AI models and what they do:** the main families of models and where each is used:
   - Speech-to-text and text-to-speech
   - Local language models and text recognition from images (OCR)
   - Image and video models, including detection for uses such as natural disasters
   - Generative AI
   - IoT models that act on sensor data, for example to trigger sprinklers or drones
   - Privacy-preserving models trained across many devices without sharing the data (federated learning)
   - Knowledge bases and chatbots
   - An introduction to AI agents
6. **Putting AI into service:** choosing between small models on devices and cloud-based services, AI on wearables, and the developer tools and communities that support deployment.
7. **Standards and good practice:** covered throughout the modules: common data formats, data governance and access control, safeguards on what models may do, and managing models over their life. Participants are also introduced to ITU AI resources, including open datasets, AI challenges, webinars and the AI for Good Sandbox.

By the end of this level you will be able to

- Use the key terms of AI and machine learning with confidence
- Explain each stage of building an AI system
- Recognize the main types of AI models and what each is used for
- Describe an AI use case for your own field in the standard ITU format

Level 2: Advanced Training and Use-Case Analysis

Date: Tuesday 20 October 2026

Duration: 10:00 am – 03:00 pm

Who it is for: AI specialists, technical teams and experts in vital sectors. Level 1 knowledge, or equivalent experience, is recommended.

This level is practical. Participants work in a training version of the AI for Good Sandbox, using low-code tools, to analyze, build and test AI solutions for their own sector.

Part A: General session (3 hours)

1. **The AI for Good Sandbox:** how the Sandbox works, with case studies of AI in different sectors. Covers the role of open data, open-source tools, research, standards and infrastructure in building trusted AI solutions.
2. **Data, privacy and trust:** finding and managing data sources, licensing and data-sharing agreements, measuring fairness and bias in AI, privacy and security safeguards, personal data protection standards and data governance.
3. **Generative AI content:** managing hallucinations and harmful content, detecting fakes, and monitoring and responding to content risks.
4. **Local and regional AI:** developing AI that reflects local language, culture and knowledge; adapting models through fine-tuning, prompt-tuning and evaluation; and scaling local solutions to other regions.
5. **Integrating AI into work processes:** measuring gains in automation, efficiency, service quality, cost and processing time. Also covers how people work with and supervise AI, how to measure trust in AI outputs, and how open and closed models compare.
6. **Building and testing AI agents:** the technical foundations of AI agents:
   - Adapting models through fine-tuning and prompt design
   - Connecting agents to tools and systems through APIs
   - Retrieval-augmented generation (RAG) and embedding models
   - Data pipelines for agents, with logging, audit trails and support for several languages and media
   - Testing agents in the Sandbox: benchmarking, performance checks, and scenario and test design
   - Open-source frameworks and protocols for agents to work together (A2A, MCP, ACP and ANP)
   - Coordinating agents, monitoring them over their life and optimizing their resource use
7. **Energy and infrastructure:** the energy used by AI for computing, storage and networks, and how to make AI infrastructure reliable and sustainable.
8. **Hands-on practice:** participants identify use cases, improve them with the ITU AI for Good Sandbox Agentic Toolkit, and check them against AI policies and strategies. Each participant leaves with a use case written in the ITU format and a set of policy discussion points.

Part B: Sector sessions (1 hour for each sector)

Separate sessions for telecommunications, education and healthcare, based on simulated scenarios:

- Sharing real use cases from the sector
- Analyzing each use case in the Sandbox with the trainers
- Identifying gaps in current policies and deriving inputs for national AI strategies
- Building a sector knowledge base and training a model together

By the end of this level you will be able to

- Use the AI for Good Sandbox to test "what if" scenarios
- Design AI use cases for your sector with the right technical building blocks

- Understand how AI agents are built, connected and tested
- Identify policy gaps that affect AI in your sector

Opening Ceremony and High-Level Panel

Date: Wednesday 21 October 2026

Time: 09:00 am – 4:00 pm

Who it is for: Their Excellencies or their representatives, by personal invitation

The official opening of the week, organized jointly by ITU and TDRA. The day presents the opportunities the ITU AI for Good Sandbox opens for the UAE and how government entities, researchers, academia and startups can take part. It brings together international and regional experts and decision makers to share experience on AI solutions, policies and strategies.

Program

1. **Opening session:** opening remarks by ITU and TDRA, followed by opening keynotes.
2. **Technical track:** international experience in building AI, with an expert panel:
   - Building regional large language models
   - Open knowledge bases for specific sectors
   - Agentic AI technology
   - Expert panel on national hackathons, turning research into standards and solutions, and standards gaps found through AI solutions
3. **Policy track:** the policies and strategies that enable AI solutions:
   - Strategies to enable innovative AI solutions
   - AI strategies for telecommunications, healthcare and education
   - Policy panel on priority sectors, testing and benchmarking with sandboxes, monitoring AI solutions, and the role of data, training and infrastructure in the regional AI ecosystem
4. **High-level panel:** decision makers discuss how AI strategies, solutions, standards and skills connect, and agree on global collaborations through the Sandbox.
5. **Announcements:** new initiatives hosted in the ITU AI for Good Sandbox – UAE:
   - Launch of the Sandbox Network for Arabic AI (SNAI), a pan-Arab platform linking national Arabic language models
   - Launch of the AI for Good Sandbox Global Hackathon: AI for Good Lab Hackathon: United Arab Emirates - AI for Good
   - Call for collaborative projects on open knowledge bases for specific sectors: https://aiforgood.itu.int/ai-for-good-sandbox-project-proposals-united-arab-emirates/
   - Launch of a global open repository of AI agents

What attendees will take away

- An understanding of what the AI for Good Sandbox offers the UAE and how their entity can take part
- Insight into how other countries are building AI solutions, strategies and policies
- Awareness of the hackathon and the collaborations launched during the week

Level 3: Leaders Program in Agentic AI

Date: Thursday 22 October 2026

Duration: 10:00 am – 12:00 pm

Who it is for: Leaders, decision makers and policy makers in government (Their Excellencies or their representatives)

This level is designed for leaders. It focuses on the decisions leaders make about AI rather than on how it is built. Its lens is agentic AI: AI systems that can plan and carry out multi-step tasks, which is the next stage of AI in government services. The session combines a guided demonstration of AI agents working in the Sandbox with discussion.

What you will learn

1. **What agentic AI means for government:** what AI agents are, how they differ from chatbots, and the journey from design to deployment in the public sector.
2. **AI agents in action across sectors:** real examples of where agents add value and where people must stay in control:
   - Education: tutoring, assessment and tracking student progress
   - Healthcare: patient triage that routes cases to specialists, with human oversight required
   - Public services: handling permit applications and escalating to officers for final approval
   - Logistics: procurement, compliance, demand forecasting, and fleet and route planning
   - Telecommunications: predicting network failures and training engineers
3. **Governing AI agents:** the policies and safeguards that make agents safe to use: guardrails and how they are enforced; national, organizational and sector guidelines; rules on which tools and data agents may use; explainability and transparency of agent decisions; bias in the sources agents consult; resolving conflicts; and keeping humans in the loop.
4. **Sovereignty and accountability:** keeping data and AI under national control, making agent actions traceable through logging and audit, and serving users in Arabic and other languages.
5. **From strategy to implementation:** coordinating AI policy across levels of government, measuring whether policies are working on services, data and models, turning policy intent into action, and the role of international cooperation and agreements.
6. **Building national capability:** developing AI skills for specialists, sector experts and everyday users; growing locally developed AI; using AI for social inclusion and accessibility; and the energy demands of AI infrastructure.
7. **Why standards and sandboxes matter:** how shared standards let agents work together across entities, and how testing in a sandbox gives assurance before a service goes live.

By the end of this level you will be able to

- Explain what agentic AI can and cannot do for government
- Identify where AI agents could improve services in your entity
- Ask the right governance questions before approving an AI agent
- Understand how the Sandbox supports safe adoption and evidence-based AI policy

What participants receive

- Follow-up support
- Access to the ITU AI for Good Sandbox, with hands-on support, open-source models and cloud credits for practical projects

- Opportunities to publish in the ITU Journal on Future and Evolving Technologies (JFET) and present in the AI for Good Global Summit in Geneva 2027.
- A chance to contribute to AI policy and strategy by identifying gaps in current policies, strategies and standards

Note: The programme is provisional and may be updated once the detailed agenda is finalised.